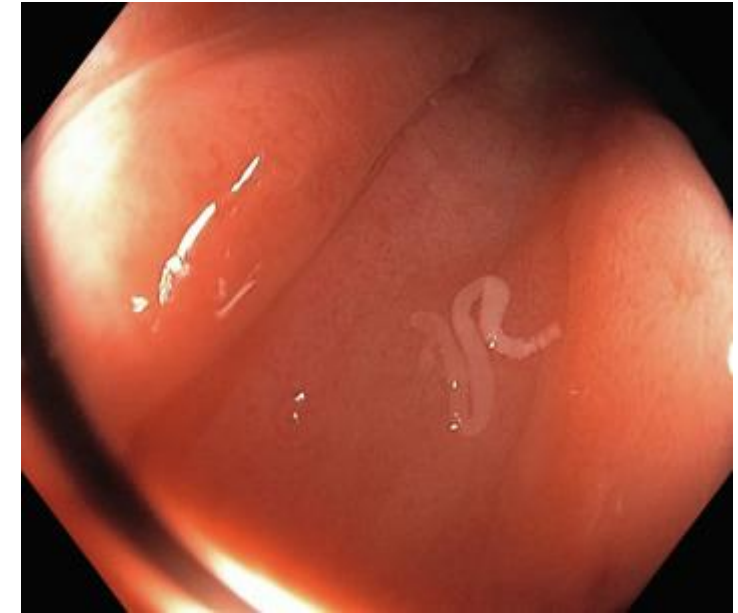
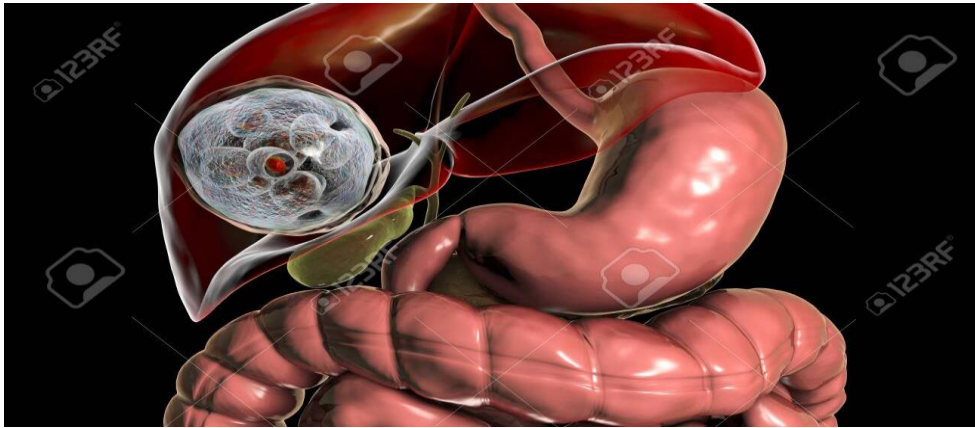


Echinococcus granulosus, and *Hymenolepis nana*



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Echinococcus granulosus



- History

Common name: (Dog tape worm)

Hydatid cysts had been described by Hippocrates and other ancient physicians.

Adult *E. granulosus* was described by Hartmann in the small intestine of dog in 1695 and the larval form (hydatid cysts) was recognized in 1782 by Goeze.



Echinococcus granulosus (the small, white objects) in the small intestine of a dog.

- **Distribution**

- Echinococcus granulosus (dog tape worm) causes **echinococcosis or hydatid disease**.
- Cestode parasite.
- The disease is prevalent in most parts of the world, it is most extensive in the **sheep and cattle** raising areas of Australia, Africa, and South America.
- It is also common in Europe, China, and the Middle East.
- It is a significant health problem in India. It is seen more often in temperate than in tropical regions.

Echinococcus granulosus

- **Habitat**
- The **adult worm** lives in the jejunum and duodenum of **dogs** and other canine carnivora (wolf, fox).
- The **larval stage** (hydatid cyst) is found in **humans and herbivores animals** (sheep, goat, cattle and horse).
- The larva of *E. granulosus* causes unilocular hydatid cyst disease (**Cystic hydatid disease**).

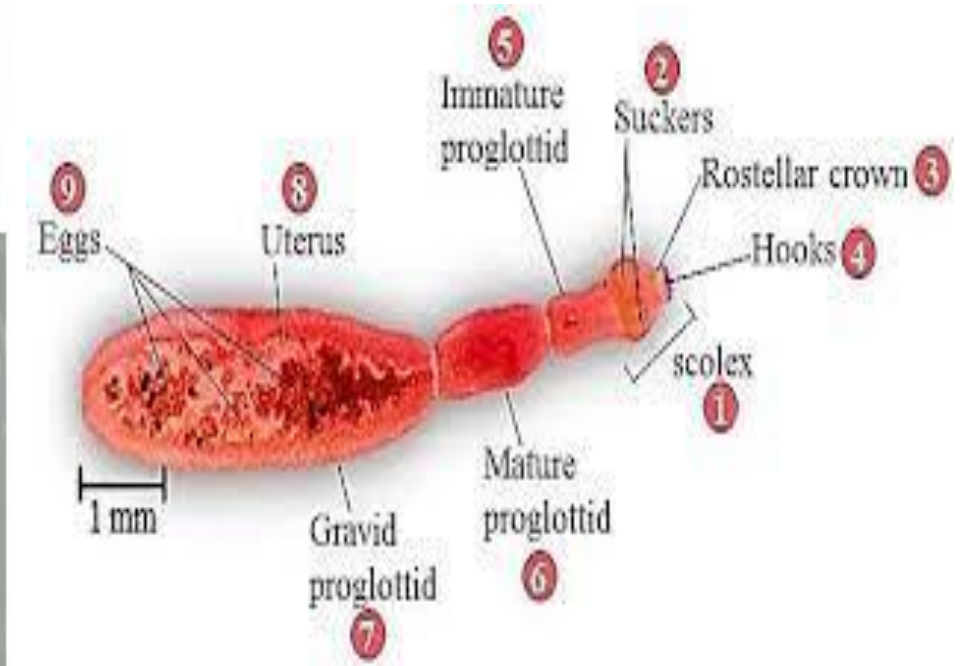


Adult of *Echinococcus granulosus*

Hydatid Tapeworm

- Scolex, neck, 3 segments
- Egg , hexacanth oncosphere

- **Largest larval stage of all tapeworms:**
Hydatid cyst



- The adult worm lives for 6–30 months.

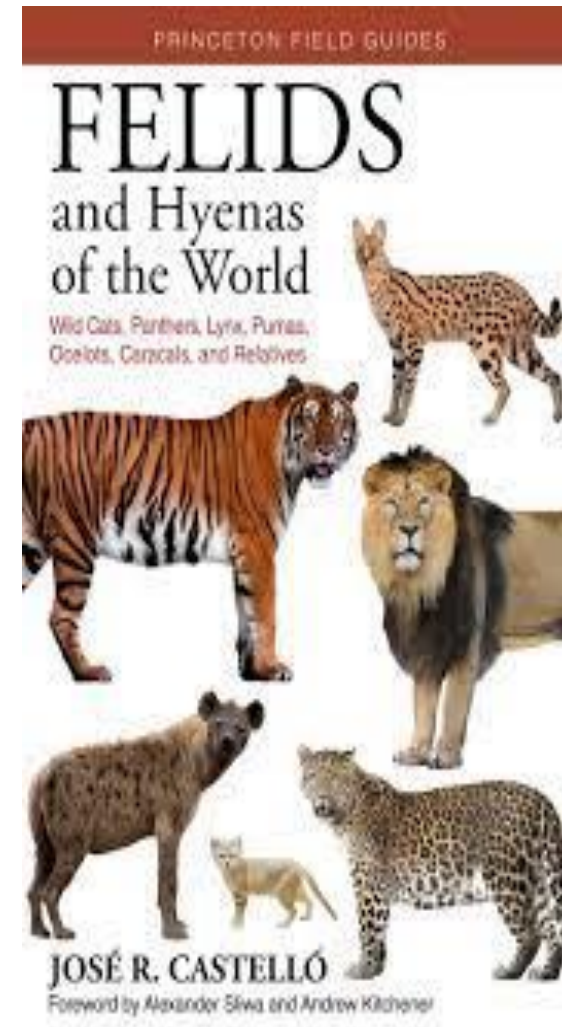
Zoonotic Transmission

Definitive hosts

- Canids
- Felids
- Hyenids

Intermediate hosts

- Herbivores
 - Humans
- Humans act as intermediate hosts
- Ingest tapeworm eggs shed by definitive hosts.
- Contaminated fruits, vegetables, herbs, water





In endemic areas **dogs** often have ready access to viscera from slaughtered **livestock** as seen in this picture



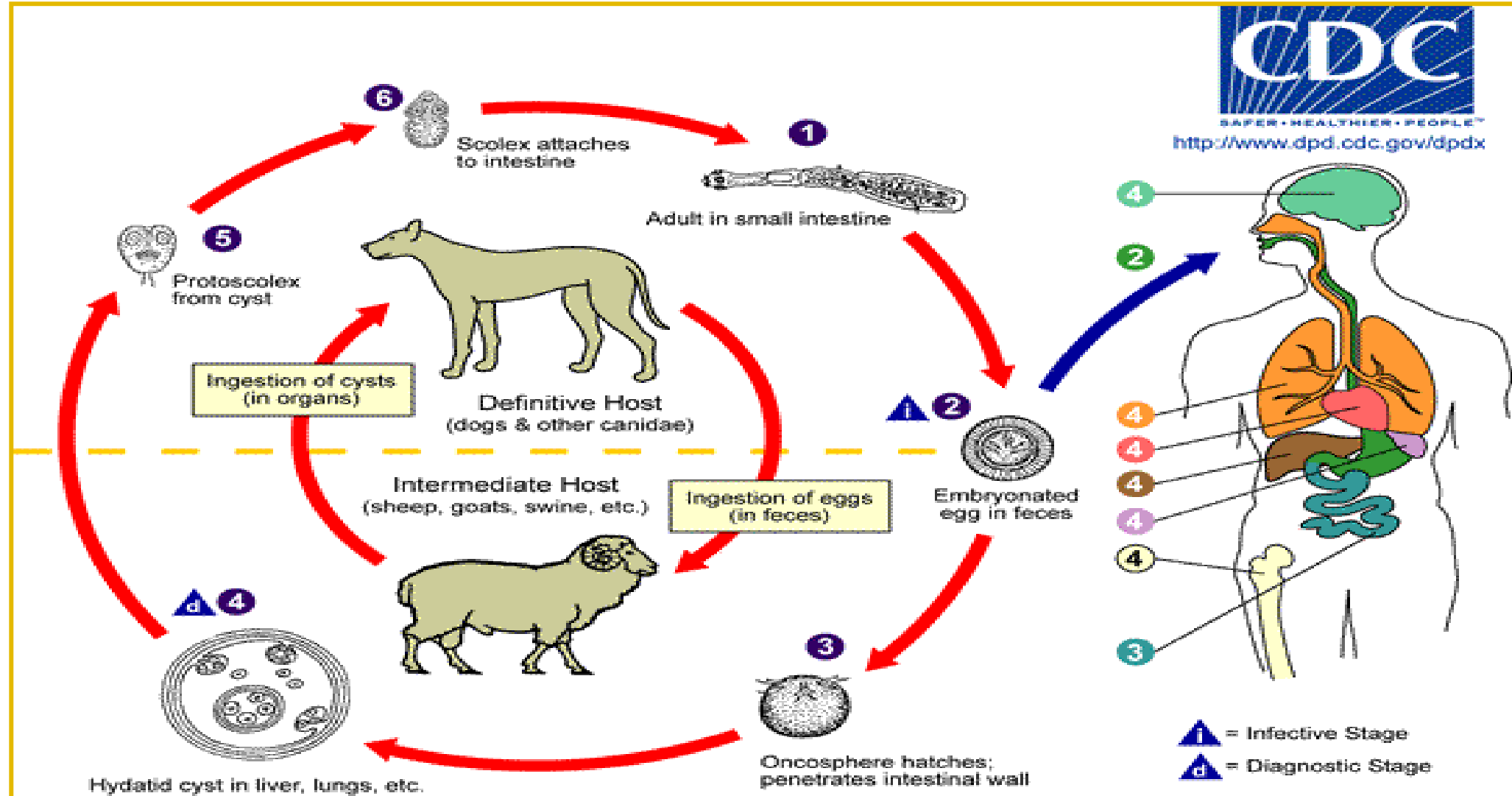
Transmission of cystic hydatid disease
Sheep slaughtering close to **free-r**
anging dogs.

Life cycle

- Final host: **carnivores**
- Intermediate hosts: **Herbivores and Human**
- When sheep or cattle harboring hydatid cysts die or are slaughtered, dogs may feed on the carcass or offal. Inside the intestine of dogs, the **scolices** develop into the **adult worms** that mature in about 6–7 weeks and produce eggs to repeat the life cycle.
- When infection occurs in humans accidentally, the cycle comes to a **dead end** because the human hydatid cysts are unlikely to be eaten by dogs.

Life Cycle

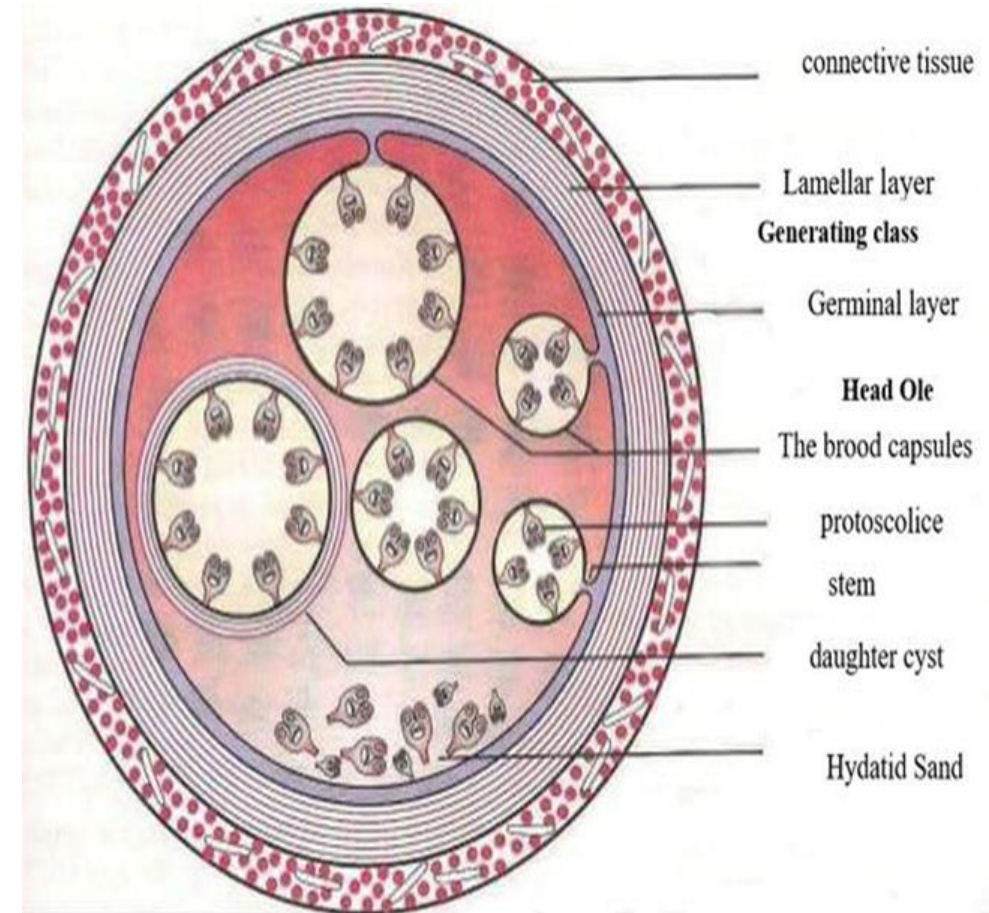
The worm completes its life cycle in 2 hosts:



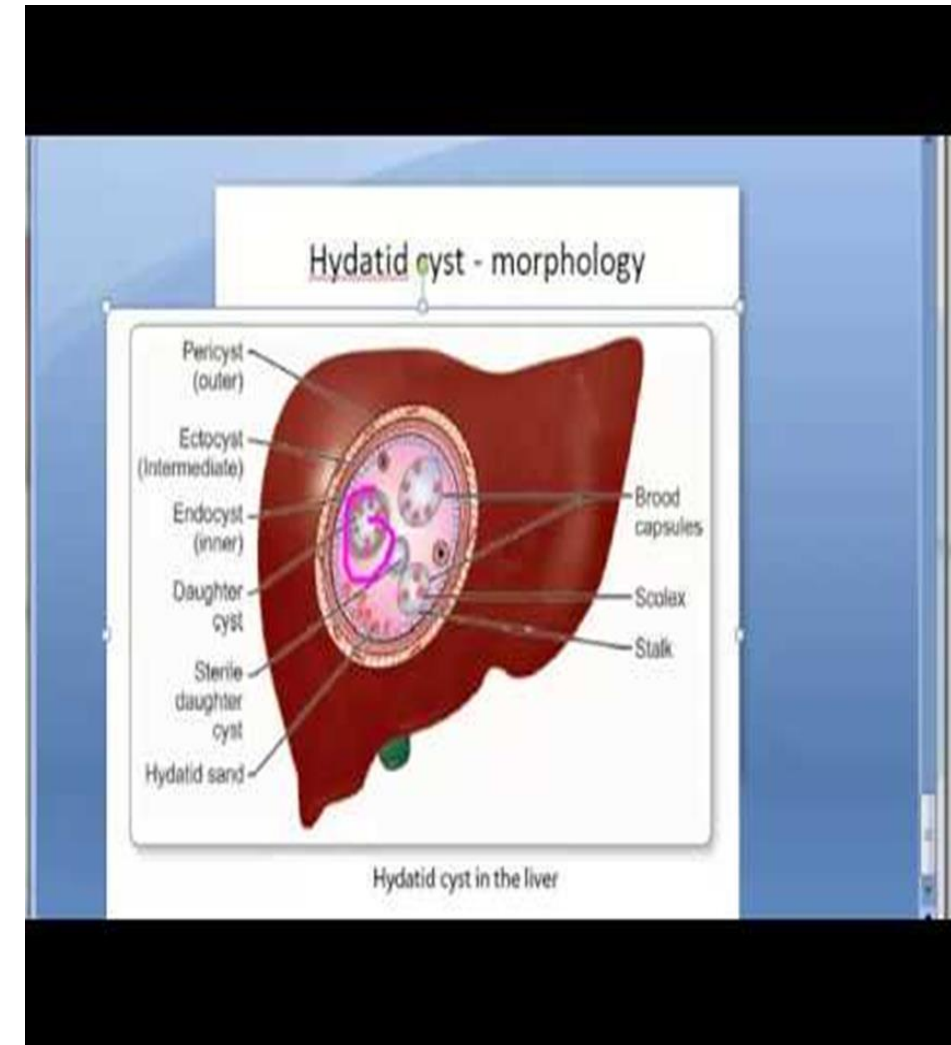
Pathogenesis

- Evolution of Hydatid Cyst

- At the site of deposition, the embryo slowly develops into a hollow bladder or cyst filled with **fluid**. This becomes the hydatid cyst (Greek hydatis: a drop of water). that contains thousands of individual **scolices** as well as many daughter cysts within the large cyst. Individual scolices lying at the bottom of the large cyst are called “**hydatid sand**.”

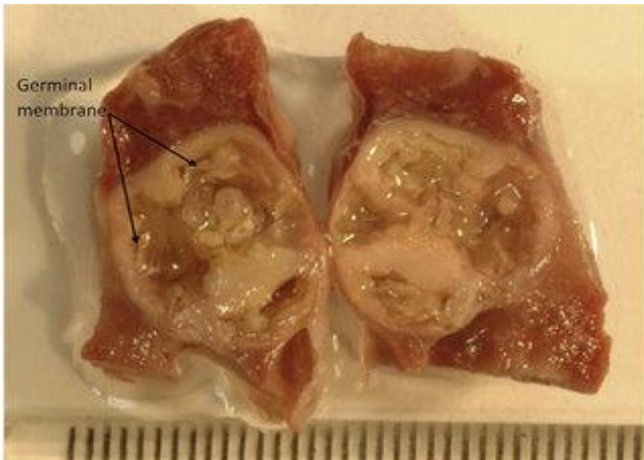


- The hydatid cyst enlarges slowly and reaches a diameter of 0.5–1 cm in about 6 months. The growing cyst evokes host tissue reaction leading to the deposition of fibrous capsule around it. The cyst fluid contains parasite antigens, which can sensitize the host. Later, if the cyst ruptures spontaneously or during trauma or surgical removal, life-threatening **anaphylactic shock** can occur. Rupture of a cyst can also spread **protoscolices** widely.

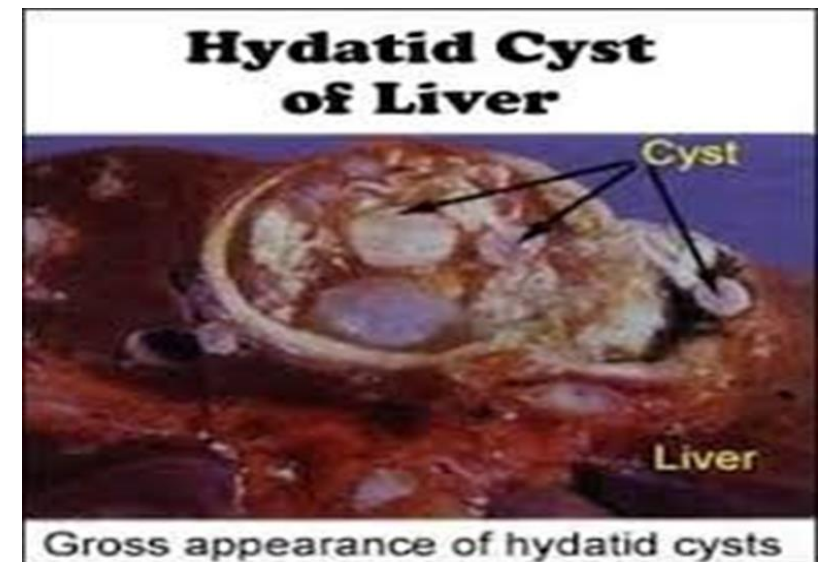
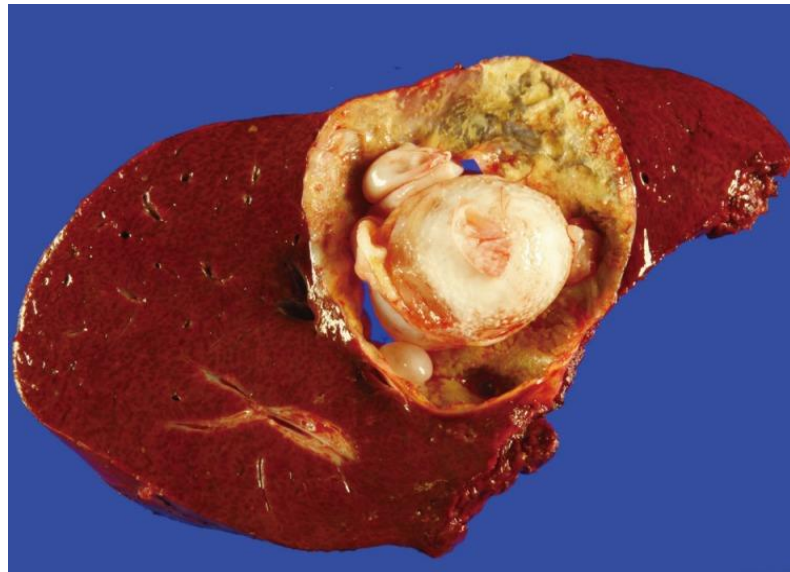


The Hydatid Tapeworm **Clinical Manifestations**

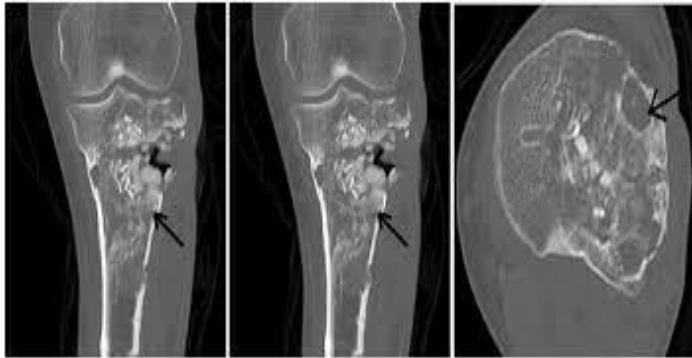
- About **75% of hydatid cyst develop in liver**, which acts as **the first filter for embryo**. However, some embryo which pass through the liver, enter the right side of heart and are caught in pulmonary capillaries (forming pulmonary hydatid cysts), so that the **lung acts as the second filter**.



Pulmonary hydatid cyst



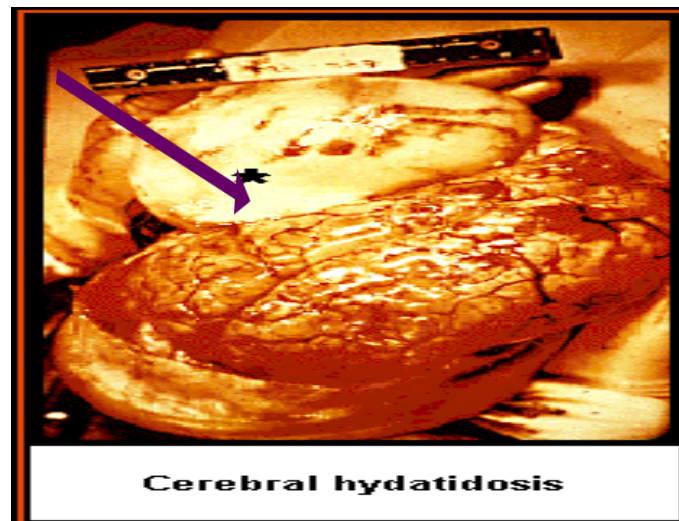
- A few enter the systemic circulation and get lodged in various other organs and tissues such as the spleen, kidneys, eyes, brain, or bones.



Hydatid cyst in knee joint



Ocular Hydatid cyst



Cerebral hydatidosis



Renal Hydatid cyst

The Hydatid Tapeworm Clinical Manifestations

- Incubation period month to years (20 to 30 years documented for cysts that grow slowly).
- Most of the times infection is **asymptomatic** and accidentally discovered.
- Clinical disease develops only when the hydatid cyst has grown big enough to cause obstructive symptoms. Disease results mainly from pressure effects caused by the enlarging cysts.
- Symptoms dependent on cyst location (60 to 70% in liver, and 20 to 25% in lungs).
- The clinical manifestations of this infection therefore vary greatly, depending on the site and size of the cyst, but resemble those of a slow-growing tumor that causes gradually increasing pressure.

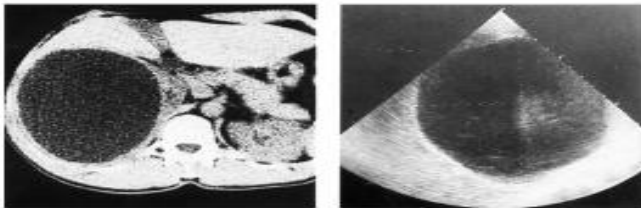
- hydatids produce many other symptoms, include collapse of infected long bones, blindness, and epileptic seizures.
- Liver cysts cause obstructive jaundice.
- Bronchial cysts may produce pulmonary abscess.
- Brain cysts produce intracranial pressure and sometimes epilepsy.
- Kidney cysts cause renal dysfunction.
- The rupture of a hydatid cyst may induce sudden anaphylactic shock in a previously asymptomatic individual.

Diagnosis

- Imaging techniques

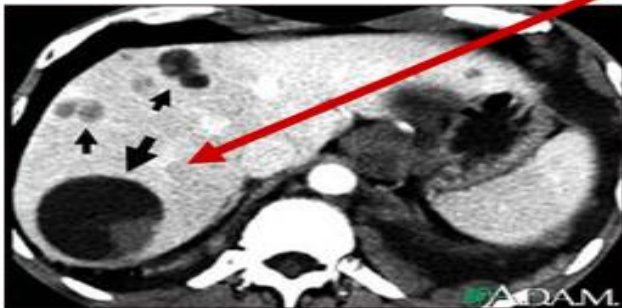
- Radiological examinations and other imaging techniques such as **ultrasonography (USG)**,

CT scan, and MRI reveal the diagnosis in most cases of cystic echinococcosis.

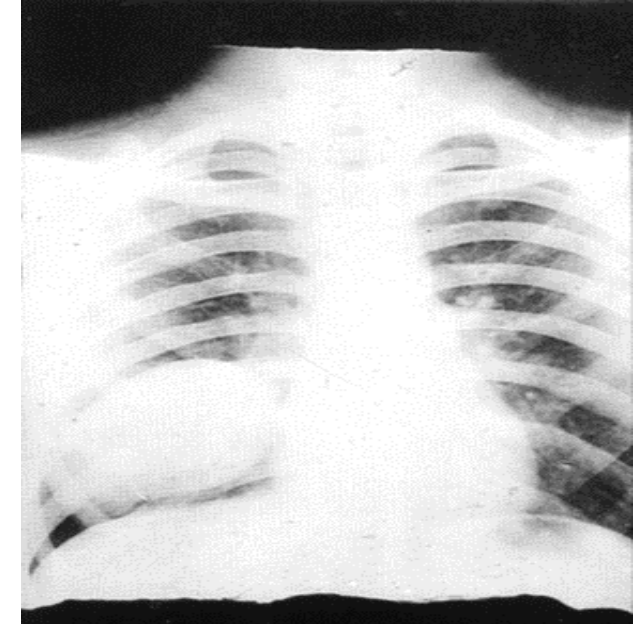


This upper abdominal CT scan shows multiple cysts in the liver, caused by Echinococcus.

Note the large circular cyst (seen on the left side of the screen) and multiple smaller cysts throughout the liver.



<http://www.drkoop.com/ency/93/ImagePages/1177.html>

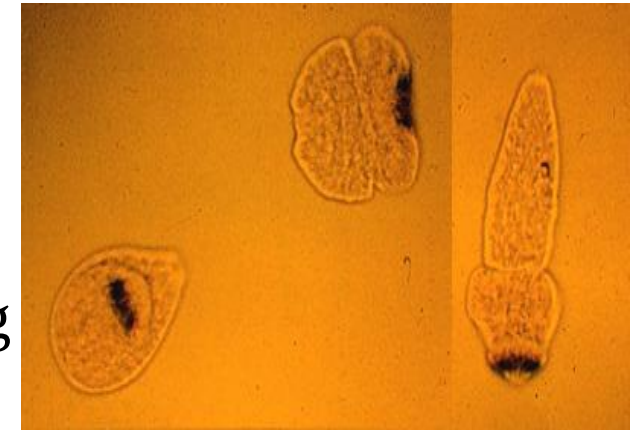


Radiology: pulmonary

Diagnosis



- Serology
- Antigen detection Specific echinococcal antigen in sera and in CSF can be detected by counter immunoelectrophoresis technique (CIEP)
- Blood Examination It may reveal a generalized eosinophilia of 20–25%
- Detection of protoscolices (Urine, sputum):
Excretion of scolices into the sputum or urine can be demonstrated by lactophenol cotton blue (LPCB) staining
- Molecular diagnosis (PCR).



Fluid aspirated from a hydatid cyst will show multiple protoscolices (size approximately 100 μm), each of which has typical hooklets.



- Prevent farm dogs from eating the offal (internal organs) of dead livestock



- Wash hands, especially after being with animals

PREVENTION



Burn carcasses of dead livestock

- Don't let a dog lick your face



Treatment

- In early stages, the treatment of choice is puncture, aspiration, injection, and reaspiration (PAIR).
- **Surgery** it is the treatment of choice for complicated E. granulosus cysts (Care must be taken not to rupture, the cyst as any fragment of the germinal layer can grow into a new cyst).
- **Fluid within the cyst can also cause anaphylaxis .**
- Recurrence after surgery is common.
- Pre and post-operative chemotherapy with albendazole after curative surgery is recommended.

Hymenolepis nana

- Common name: Dwarf tape worm
- History and Distribution
- The name Hymenolepis refers to the thin membrane covering the egg (Greek hymen—membrane, lepis—rind or covering) and nana to its small size (nanus—dwarf).
- It was first discovered by Bilharz in 1857.
- It is cosmopolitan in distribution but is more common in warm than in cold climates.
- Infection is most common in school children and institutional populations.

Hymenolepis nana

Hymenolepis nana is the **smallest** and the most common tapeworm found in the **human intestine**.

It is unique that it is the only cestode which completes its life cycle in one host—**humans**



Hymenolepis nana

Adult Worm

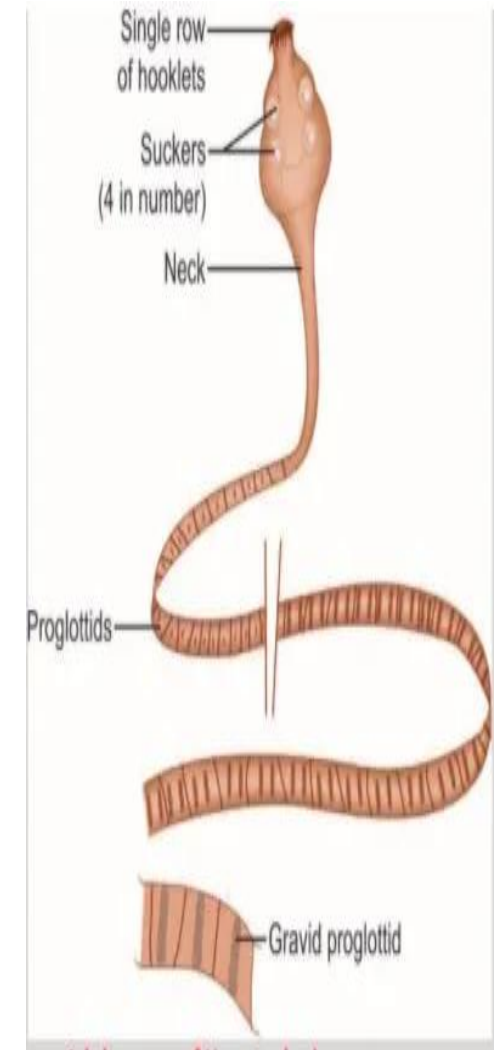


H. nana is the **smallest intestinal cestode** that infects man.

The adult worm lives in the ileum of man. It is 5–45 mm in length and less than 1 mm thick.

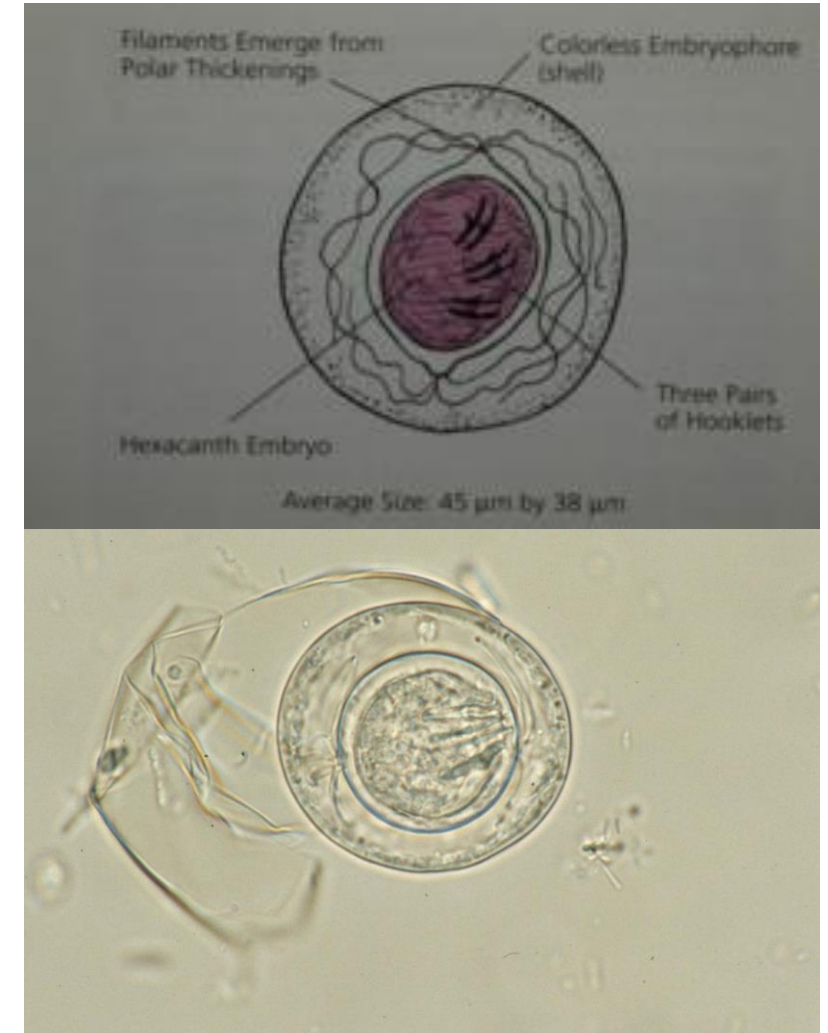
The scolex has 4 suckers and a retractile rostellum with a single row of hooklets

The long slender neck is followed by the strobila consisting of 200 or more proglottids, which are much broader than long.



Hymenolepis nana egg

- egg is roughly spherical or ovoid, 30–40 μm in size.
- It has a thin colorless outer membrane and
- inner embryophore enclosing the hexacanth oncosphere.
- The space between 2 membranes contains yolk granules and 4–8 thread like polar filaments
- The eggs are **immediately infective** and unable to survive for more than 10 days in external environment



(life cycle)

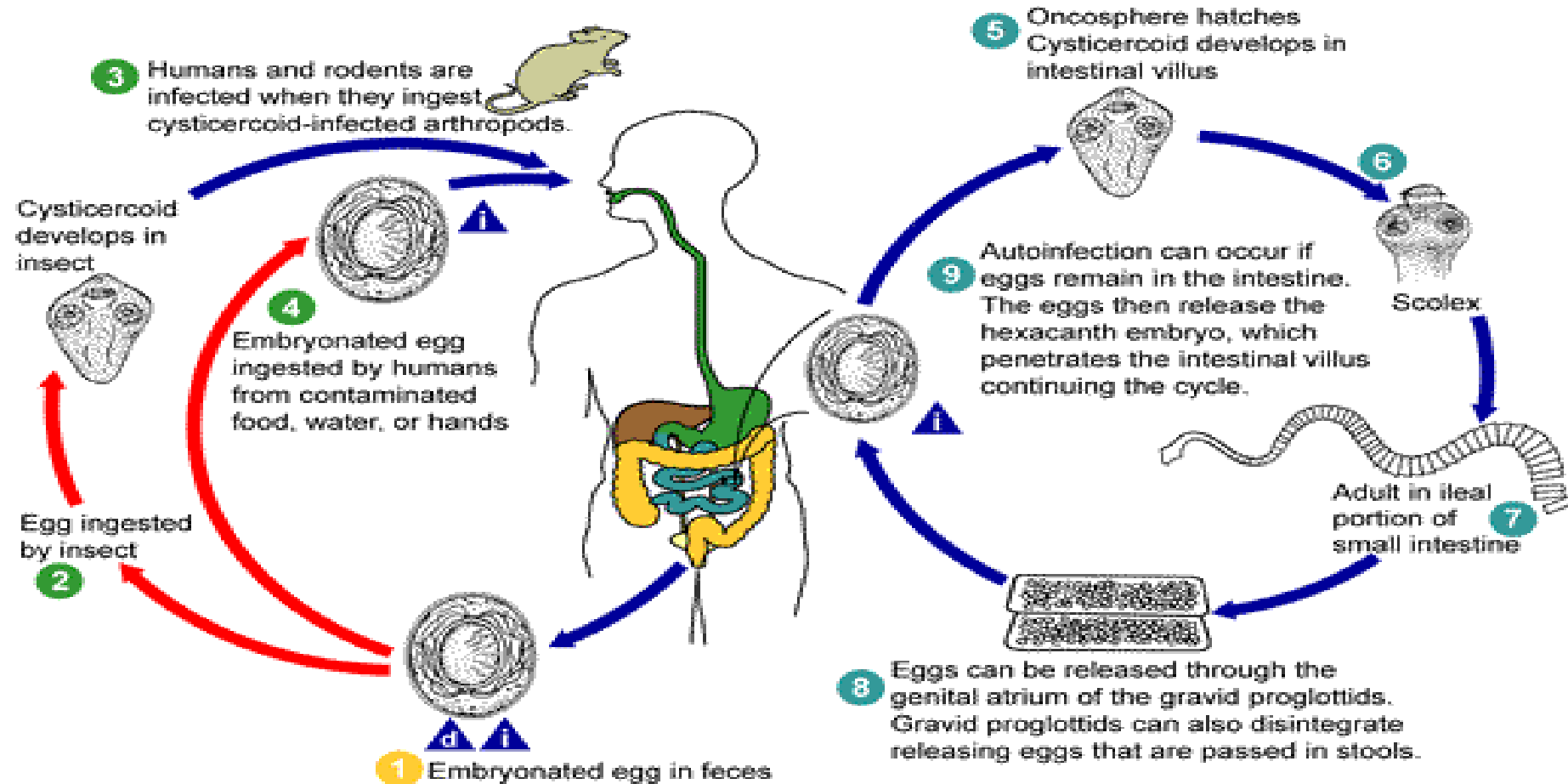
Host: Man.

There is no intermediate host.

i = Infective Stage
d = Diagnostic Stage



<http://www.dpd.cdc.gov/dpdx>



Clinical Features

- Hymenolepiasis occurs more commonly in children.
- There are usually no symptoms but in heavy infections, there is nausea, anorexia, abdominal pain, diarrhea, and irritability.
- Sometimes pruritus may occur due to an allergic response.

• Prophylaxis

- Maintenance of good personal hygiene and sanitary improvements.
- Avoiding of consumption of contaminated food and water.
- Rodent control.

Diagnosis

- The diagnosis is made by demonstration of characteristic eggs in feces by direct microscopy.
- ELISA test

• Treatment

- Praziquantel is the drug of choice, since it acts both against the adult worms and the cysticercoids in the intestinal villi.

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